

Project Design

Problem Solution

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Project Name	Automated Network Request Management in ServiceNow

1. Customer Segment(s) [CS]

Primary Requesters: Individual corporate employees, project engineers, developers, and academic staff/students who need dedicated network connectivity, bandwidth allocations, or port reconfigurations to execute daily tasks.

Delegated Submitters: Administrative coordinators or team assistants utilizing the 'Opened on behalf of' field to request setup for new hires, visiting engineers, or contract resources.

Approving Authorities: Line managers, department heads, and network security compliance officers responsible for validating access permissions, hardware readiness, and cost allocations.

2. Jobs-To-Be-Done & Problems [J&P]

Core Jobs-To-Be-Done: Submit network connection specifications quickly through an intuitive self-service portal, dynamically toggle between 'New' and 'Existing' connections, select valid payment channels (UPI/Card), accurately compute total billing/tariff amounts, and provide exact on-premise installation addresses.

Key Friction Points: Prior workflows forced users to navigate disjointed forms, fill redundant contact information, guess approval chains, and repeatedly follow up via email or chat to determine if access was granted.

3. Triggers [TR]

Operational Onboarding: New employee workstation setup, branch office onboarding, or laboratory workstation deployment requiring immediate LAN/VLAN connectivity.

Infrastructure Modifications: Upgrading connection speeds, reassigning existing network drop IDs, or relocating desks within an office campus.

Impending Deliverables: Urgent project kickoff deadlines where network access is a blocking dependency for software testing and environment connectivity.

4. Emotional Journey: Before vs. After [EM]

Before Implementation (Negative State): Anxious about project schedule slippage; frustrated by repetitive manual forms; overwhelmed by lack of visibility into approval ownership; stressed by unacknowledged email tickets.

After Implementation (Positive State): Reassured and confident; in complete control of request tracking; relieved by receiving instantaneous automated email confirmations and milestone updates.

5. Existing Solutions & Limitations [AS]

Legacy Channels: Manual email threads to IT helpdesks, paper-based requisition slips, and generic incident tickets in shared inboxes.

Systemic Drawbacks: High manual triage overhead, lack of standardized inputs leading to repeated re-submissions, missing technical attachments, zero SLA enforcement, and complete absence of audit trails for compliance.

6. Customer Constraints [CC]

Technical Constraints: End users have zero network engineering expertise and cannot diagnose port types, VLAN IDs, or subnet routing independently.

Security & Policy Constraints: Mandatory corporate network compliance, requiring rigorous manager sign-off prior to network port opening.

Financial Constraints: Clear budget allocation or immediate payment confirmation (UPI/Card) required before physical provisioning commences.

7. User Behaviour [BE]

Intake Navigation: Logs into ServiceNow Service Portal (/sp), searches for 'Network Request', and accesses the clean catalog item form.

Guided Data Entry: Verifies pre-filled contact details, selects connection type, provides mandatory address and payment information, and attaches network layout schematics.

Status Monitoring: Monitors personal email inbox for automated receipts, approval notices, and ticket resolution records without opening support follow-ups.

8. Channels of Behaviour [CH]

Online Digital Touchpoints: ServiceNow Service Portal (/sp) self-service interface, automated outbound transactional SMTP notifications, and personal request status lists.

Offline & Physical Touchpoints: Internal manager briefings regarding justification, and subsequent physical workstation or router installation executed by network engineers.

9. Problem Root Cause Analysis [RC]

Absence of Integrated Intake: Legacy systems lacked a centralized digital portal item tailored specifically for network connectivity requests.

Missing Client-Side Controls: Absence of dynamic UI policies resulted in either excessive cluttered fields or missing critical parameters (e.g., omitting 'Existing ID' during existing line modifications).

Fragmented Server Automation: Disjointed handoffs between request logging and approval routing, with no automated persistence table or workflow engine to enforce lifecycle states.

10. Proposed Solution Architecture [SL]

Service Portal Form Interface: A streamlined 'Network Request' catalog item featuring dynamic UI Policies. Automatically enforces mandatory fields, selectively displays 'Existing ID' only when Type of Connection is 'Existing', and standardizes payment methods (UPI/Card).

Database Tables Persistence: Automated capture of all submitted parameters into a dedicated custom table: Database Tables [u_database_table], featuring auto-numbering (DBT0001001) and audit logging.

Flow Designer 10-Step Workflow: An end-to-end asynchronous workflow:

- Trigger: Service Catalog submission.
- Step 1: Get Catalog Variables from Network Request.
- Step 2: Create Record in Database Tables [u_database_table].
- Step 3: Dispatch Submission Acknowledgment Email to requester.
- Step 4: Ask For Approval on Database Tables record targeting designated manager.
- Step 5: Evaluate If Request is Approved condition.
- True Branch (Steps 6-7): Update Database Tables record to 'Approved' and send approval email.
- False Branch (Steps 9-10): Update Database Tables record to Rejected' and send rejection email.